# MACHINE LEARNING

## Machine Learning

Machine Learning (ML) enables computers to learn and make predictions without explicit programming. It is used in areas like image and speech recognition, translation, and recommendation systems (e.g., Netflix – movie recommendations).

## Need for Machine Learning

Traditional programming fails to handle complex data and large volumes. ML helps by learning from examples, automating tasks, and improving over time (e.g., spam detection, fraud detection, voice assistants).

## Importance of Data

Quality data is essential for accurate predictions. Diverse and sufficient datasets improve performance, generalization, and real-world adaptability.

## Types of Machine Learning

1. Supervised Learning – Learns from labeled data for classification or regression (e.g., disease prediction).
2. Unsupervised Learning – Works on unlabeled data to find patterns or clusters (e.g., customer segmentation).
3. Reinforcement Learning – Learns by trial and error with rewards and penalties (e.g., selfdriving cars, games).

## Supervised Machine Learning

Uses labeled data where both input and output are known. Model learns patterns to predict new outcomes. Used in spam detection, stock prediction, medical diagnosis, etc.

### Common Algorithms

Linear Regression – Predicts continuous values.

Logistic Regression – Binary classification (Yes/No).

Decision Trees – Splits data into branches for decisions.

Random Forest – Combines many trees for accuracy.

SVM – Finds best hyperplane for classification.

KNN – Classifies based on nearest data points.

## Unsupervised Learning

Works on unlabeled data to discover hidden structures or relations (e.g., clustering customers, anomaly detection).

### Main Algorithms

1. Clustering – Groups data (K-means, Hierarchical, DBSCAN).
2. Association – Finds relations (Apriori, FP-Growth – Market Basket Analysis).
3. Dimensionality Reduction – Reduces features (PCA, LDA).
4.

### Applications & Challenges

Used in segmentation, recommendation, fraud detection. Challenges include noisy data, parameter tuning, and difficulty in result interpretation.

## Predictor Variable

Predictor (independent) variable helps forecast or explain the dependent variable. Examples: study hours → exam score, age → disease risk.

Types – Continuous (age), Categorical (gender), Binary (yes/no), Ordinal (ranked levels).

## Explanatory vs Response Variables

Explanatory (independent) variables explain outcomes; Response (dependent) variables show results. Example: dosage → recovery rate (cause and effect).

## Classification

Classification assigns data to categories (e.g., spam/not spam). Types: Binary (2 classes), Multiclass (3+ classes), Multi-label (multiple categories at once).

Applications: Spam detection, medical diagnosis, sentiment analysis, image recognition, fraud detection.

## Regression

Regression predicts continuous numerical outcomes (e.g., predicting house price). Types include Simple, Multiple, Polynomial, Ridge, Lasso, SVR, Decision Tree, and Random Forest regression.

Evaluation Metrics – MAE, MSE, RMSE, $R^2$ Score.

Applications – Price prediction, trend forecasting, risk analysis.

## LINEAR REGRESSION

### Linear Regression

Linear Regression is a supervised machine learning algorithm used to predict continuous numerical values.
It establishes a linear relationship between independent variables and a dependent variable using a best-fit line.

### Types of linear regression

- Simple linear regression
- Multi linear regression

### Why did you use train-test split?

To divide the dataset into:

- Training data (to train the model)
- Testing data (to evaluate performance)

This prevents overfitting and checks real-world performance.

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### Why did you use StandardScaler?

Because features have different ranges.

StandardScaler normalizes the data so that:

**Z= x-mean/ standard deviation**

That means:

Subtract the mean of the column. Divide by standard diviation of that column.

After scaling:

- Mean = 0
- Standard deviation = 1

This improves model stability and performance.

### When to use Standard Scalar

Works well for algorithms that depend on distances or gradients, such as:

- Linear Regression
- Logistic Regression
- K-Nearest Neighbour (KNN)
- Support Vector Machines (SVM)
- Principal Component Analysis (PCA)
- Neural Network (NN)

### What is the purpose of the correlation heatmap?

The heatmap shows the relationship between
features. It helps:

- Identify strongly correlated features

Detect multicollinearity

### What is the meaning of intercept?

Intercept is the predicted value of the target variable when all independent variables are zero.

### What do coefficients represent?

Coefficients show the impact of each feature on the target variable.

Positive coefficient → increases house value

Negative coefficient → decreases house value

### Mean Squared Error (MSE)

MSE measures the average squared difference between actual and predicted values.

Lower MSE indicates better performance It penalizes large errors more than small ones.

### $R^2$ Score

$R^2$ Score measures how well the model explains the variance in the data.

Range:

- 0 → Poor model
- 1 → Perfect model

Higher $R^2$ means better model fit.

### Why did you plot Actual vs Predicted values

To visually check model accuracy.

If points lie close to the diagonal line, predictions are good.

### What are the assumptions of Linear Regression

1. Linearity
2. Independence
3. No Multicollinearity
4. Homoscedasticity
5. Normal distribution of errors

RANDOM FOREST

### Ensemble Learning

Combines multiple weak models for better accuracy.

Types: Bagging (reduces variance), Boosting (reduces bias), Stacking (uses meta-model).

### Random Forest Algorithm

An ensemble of many decision trees trained on random subsets of data and features. Uses voting (classification) or averaging (regression) for prediction. Reduces overfitting and improves accuracy (e.g., used in finance, healthcare).

Advantages: Handles missing data, shows feature importance, works on large datasets.

Assumptions: Each tree learns independently; more diverse data improves accuracy.

## How Random Forest Works

1. From the original dataset, it creates **multiple random samples** (Bootstrap sampling).
2. For each sample, a **Decision Tree** is built.
3. At each split, it selects a **random subset of features**.
4. Final Prediction:
    - For **Classification** → Majority Voting
    - For **Regression** → Average of outputs

## Types of Ensemble method

- Bagging
- Boosting

## Bagging

Train multiple models **independently (parallel)** on different random samples of data.

**Purpose:**

- Reduces variance
- Prevents overfitting

Eg) Random Forest

## Boosting

Train models **sequentially**.
Each new model corrects the mistakes of the previous one.

**Purpose:**

- Reduces bias
- Improves weak learners

Eg) XGBoost

### What happens if we increase n_estimators?

- Accuracy usually improves.
- Overfitting reduces.
- Training time increases.
- After a point, improvement becomes small.

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